1.1.1. CONCEPTION PHASE:
- Task: A Use-Case of booking a hotel room: Airbnb

1: Create a requirement specification for Airbnb:
- Below mentioned is the requirement specification for the Airbnb project.
- Roles(person/user groups)

a. Guests- Customer or group of people wanting to rent out accommodations for given time-duration.
b. Hosts- Owners or managers of properties presenting accommodations on rent.
c. Administrators- Owners or managers of Airbnb platform that ensures adherence to rules and agreements, resolving conflicts and maintaining system functionality.

- Actions Performed by Roles:

a. Guests: Find listings, reservations of accommodations, exchange of communications with hosts using messaging feature, leaving feedback in the form of reviews and ratings post-stay, having ability to save suitable payment options and applying them for making payments, obeying the mentioned rules and regulations.
b. Hosts: Adding properties with proper photos, prices and descriptions, handling bookings, handling guest communication, properties maintenance, enhancing accommodation experience according to the review and rating, having the ability to save payment methods in relevance and utilising them for drawing their payouts, adhering to the mentioned rules and policies.
c. Administrators: Managing platform activity, enforcing compliance with community standards and legal specifications, resolving conflicts like refunds, penalties and cancellations.

- Required Data and Functions:
- Data Requirements:

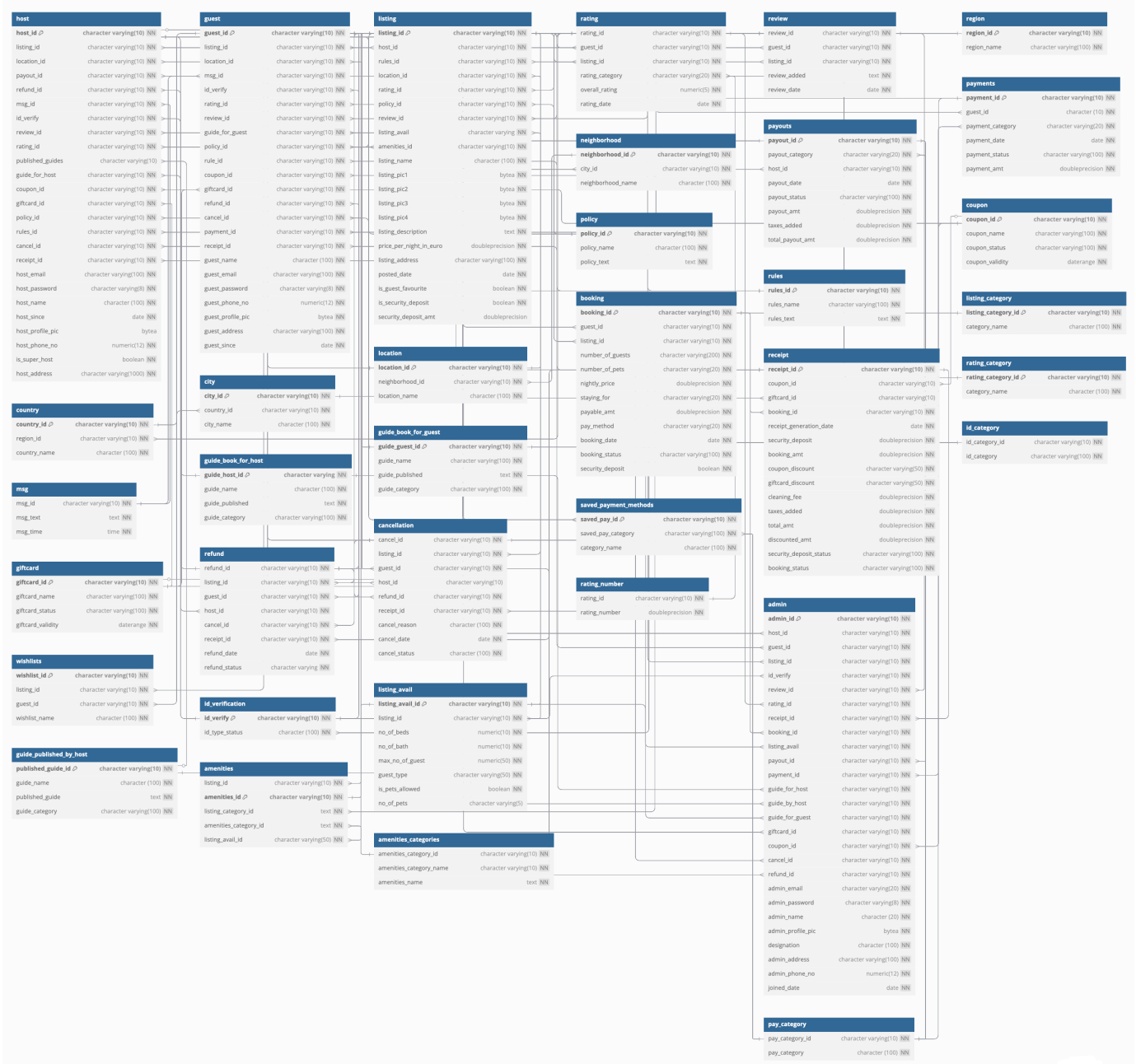
a. Guest Data: Personal details[name, email, phone number, profile photo, address], payment information, booking history, wish-lists, coupons, gift-cards, tutorial and information guides, reviews and ratings, messages with hosts, is-a-verified-guest.
b. Host Data: Personal details[name, email, phone number, profile photo, address], payment information, listings posted, coupons, gift-cards, tutorial and information guides, reviews and ratings, messages with guests, is-a-verified-host.
c. Listing Data: Property details[name, rental pictures, description, address, amenities, category], pricing structure, availability calendar, is-a-guest-favourite, reviews and ratings received from guests, is-a-verified-listing.
d. Booking Data: Booking ID, dates, payment status, agreed-upon policies, agreed-upon rules, applied coupons and gift-cards, booking status.
e. Administrator Data: Personal details[name, email, phone number, profile photo, designation], access to different fields and duties as per designation.

- Functionality Requirements:

a. Availability Functionality: Enable guests to view listings based on their choice[by filtering location, price range, amenities, dates, number of guests, number of rooms, type of listing].
b. Booking System: Booking guests with suitable listing availability, storage of relevant payment methods, generation of receipts.
c. Messaging Platform: Enabling guest-host communication.

d. Help Centre: Developing guide books for both hosts and guests to assist them in enhancing experiences, with reference to tips, tricks and techniques for hosts and guests.
e. Administrative tools: Conflict resolution system for refunds or penalties in case of cancellations or issues arise during stays, tools for enforcing compliance[photo quality checks for listing, host verification, guest verification, ensuring relevant and purposeful reviews and ratings are published].

2: ERM FOR AIRBNB



3: Data Dictionary:

-The table below contains all datatype attributes that have been used in the ERD:

Table Name	Column Name	Data Type	Constraints	Description
Host	host_id	Character Varying(10)	NOT NULL, UNIQUE, PK	Unique Identifier of the Host
Guest	guest_name	Character(100)	NOT NULL	Name of Guests
Listing	price_per_night	Double Precision	NOT NULL	Price per night for booking this listing
Message	msg_time	Time without time-zone	NOT NULL	Time stamps for each message sent
Review	review_date	Date	NOT NULL	Date of reviews posted
Admin	admin_profile_pic	Bytea	NOT NULL	Binary data containing profile picture for the admin
Coupon	coupon_validity	Date Range	NOT NULL	Specified date-range for coupon validity
Rating	overall_rating	Numeric(5)	NOT NULL	Calculated overall rating from all 5 categories
Id Verification	verify_status	Boolean	NOT NULL	Boolean(1 for yes), (0 for no) for id verification status.

- *Below is the data dictionary for this Airbnb ERD, only a few tables are expanded here as, every table in this schema follows a similar structure.*

- Table Name: Listing:

Column Name	Data Type	Constraints	Description
listing_id	Character Varying(10)	NOT NULL, UNIQUE, PK	Unique Identifier for the listing
host_id	Character Varying(10)	NOT NULL, FK	Foreign key referencing the host who owns the listing
rules_id	Character Varying(10)	NOT NULL, FK	Foreign key referencing rules applicable to the listing
location_id	Character Varying(10)	NOT NULL, FK	Foreign key referencing location of the listing
rating_id	Character Varying(10)	NOT NULL, FK	Foreign key referencing ratings associated with listing
policy_id	Character Varying(10)	NOT NULL, FK	Foreign key referencing policies for the listing

review_id	Character Varying(10)	NOT NULL, FK	Foreign key referencing reviews for listing
listing_avail	Character Varying(10)	NOT NULL	Availability status of this listing
amenities_id	Character Varying(10)	NOT NULL, FK	Foreign key referencing amenities available for the listing
listing_name	Character(100)	NOT NULL	Name of the listing
listing_pics	Bytea	NOT NULL	Binary data containing a pictures of the listing
listing_description	Bytea	NOT NULL	Binary data containing a description of the listing
price_per_night	Double Precision	NOT NULL	Price per night for booking this listing
listing_address	Character Varying(100)	NOT NULL	Address of the listing
posted_date	Date	NOT NULL	Date when this listing was posted
is_guest_favourite	Boolean	NOT NULL	Indicates if a the listing is a guest favourite
is_security_deposit	Boolean	NOT NULL	Indicates if a security deposit is required
security_deposit_amt	Double Precision		Amount required as a security deposit(if applicable)

- Table Name: Message:

Column Name	Data Type	Constraints	Description
msg_id	Character Varying(10)	NOT NULL, UNIQUE, PK	Unique Identifier for each message
host_id	Character Varying(10)	NOT NULL, FK	Identifier for the host, sending or receiving messages (references host_id in the host table)
guest_id	Character Varying(10)	NOT NULL, FK	Identifier for the guest, sending or receiving messages (references guest_id in the guest table)
msg_text	Bytea	NOT NULL	Binary data containing a content of the message
msg_time	Time without time-zone	NOT NULL	Time stamps for each message sent
msg_date	Date	NOT NULL	Date stamps for each message sent

- Table Name: Listing Availability

Column Name	Data Type	Constraints	Description
listing_avail_id	Character Varying(10)	NOT NULL, UNIQUE, PK	Unique Identifier for availability record
listing_id	Character Varying(10)	NOT NULL, FK	Identifies the listing availability and what applies to it(references listing_id in the listing table)
check_in	Date	NOT NULL	The date when guest arrives
check_out	Date	NOT NULL	The date when guest leaves
no_of_beds	Numeric(20)	NOT NULL	Number of beds available
no_of_bath	Numeric(20)	NOT NULL	Number of beds available
no_of_guest	Numeric(100)	NOT NULL	Number of guests allowed
guest_type	Character Varying(20)	NOT NULL	Type of guests allowed(Adults,children)
is_pets_allowed	Boolean	NOT NULL	Can pets be included along with guests
no_of_pets	Character Varying(10)		Number of pets allowed(if applicable)

- Table Name: Receipt

Column Name	Data Type	Constraints	Description
receipt_id	Character Varying(10)	NOT NULL, UNIQUE, PK	Unique Identifier for the receipts
booking_id	Character Varying(10)	NOT NULL, FK	Represents the booking for which this receipt was generated (references the booking_id from booking table)
coupon_id	Character Varying(10)	NOT NULL, FK	If a coupon is applied to the booking (references the coupon_id from coupon table)
giftcard_id	Character Varying(10)	NOT NULL, FK	If a giftcard is applied to the booking (references the giftcard_id from giftcard table)
receipt_generation_date	Date	NOT NULL	The date on which this receipt was generated.
security_deposit	Double Precision	NOT NULL	The amount of security deposit charged while booking a listing

cleaning_fee	Double Precision	NOT NULL	The amount charged for cleaning the booked listing
taxes_added	Double Precision	NOT NULL	The taxes charged while booking a listing
total_amt	Double Precision	NOT NULL	The total amount [security_deposit, cleaning_fee, taxes, cleaning_fee] charged while booking a listing
security_deposit_status	Character Varying(100)	NOT NULL	The status of the charged security deposit

- Table Name: Cancellation

Column Name	Data Type	Constraints	Description
receipt_id	Character Varying(10)	NOT NULL, UNIQUE, PK	Unique identifier for the cancellation record.
booking_id	Character Varying(10)	NOT NULL, FK	Identifying the booking that was cancelled(references booking_id from booking table)
host_id	Character Varying(10)	NOT NULL, FK	Identifying [If applicable], the host is involved in cancellation(references host_id from host table)
refund_id	Character Varying(10)	NOT NULL, FK	If any refund is applicable, (references refund_id from refund table) [Can be NULL]
policy_id	Character Varying(10)	NOT NULL, FK	Indicates the applied cancellation policy(references from the policy table)
rules_id	Character Varying(10)	NOT NULL, FK	Indicates the applied cancellation rules(references from the policy table)
receipt_id	Character Varying(10)	NOT NULL, FK	Linking the cancellation to specified receipt(references receipt_id from the receipt table)
cancel_reason	Character (100)	NOT NULL	Reason provided for cancellation
cancel_date	Date		Date when cancellation was initiated
cancel_status	Character (100)		Current status of the cancellation

- Table Name: Coupon

Column Name	Data Type	Constraints	Description

coupon_id	Character Varying(10)	NOT NULL, UNIQUE, PK	Unique identifier for the coupon
coupon_name	Character Varying(100)	NOT NULL	Name of the coupon
coupon_status	Character Varying(100)	NOT NULL	Status of the coupon
coupon_validity	DateRange	NOT NULL	Date range for the coupon is valid

- Table Name: Payments

Column Name	Data Type	Constraints	Description
payment_id	Character Varying(10)	NOT NULL, UNIQUE, PK	Unique identifier for the payment
payment_category	Character Varying(10)	NOT NULL	Payment method used for paying (credit-card, paypal)
guest_id	Character Varying(10)	NOT NULL	Indicates which guest made the payment (references guest_id from the guest table)
payment_date	Character Varying(10)	NOT NULL	Date on which the payment was made
payment_status	Character Varying(100)	NOT NULL	Status of the payment made.
payment_amt	Double Precision	NOT NULL	Amount that is to be paid.

*Below, **listing_id** and **listing_category_id** together form the composite primary key for the **listing_categories** table*

Column Name	Data Type	Constraints	References	Description
listing_id	Character Varying(10)	PRIMARY KEY(part of composite key)	Listing Table	Unique identifier of the listing
listing_category_id	Character Varying(10)	PRIMARY KEY(part of composite key)	Listing Category Table	Unique identifier of the listing category

*Below, **guest_id** and **listing_id** together form the composite primary key for the **wishlist** table including columns **wishlist_name, wishlist_id***

Column Name	Data Type	Constraints	References	Description

wishlist_id	Character Varying(10)	NOT NULL, UNIQUE, PK	-	Unique identifier of the wishlist created.
listing_id	Character Varying(10)	PRIMARY KEY(part of composite key)	Listing Table	Unique identifier of the listing
guest_id	Character Varying(10)	PRIMARY KEY(part of composite key)	Guest Table	Unique identifier of the listing category
wishlist_name	Character(100)	NOT NULL		Name of the wishlist

4: Summary:

An in-depth research on requirement analysis was conducted, before the beginning of the database designing process, to identify user roles such as guests, hosts, administrators and their associated actions in sense of roles such as hosts posting rental accommodations, guests booking those accommodations, and administrators keeping all these activities aligned in terms of community standards.
Subsequently, using these identified entities and significant relationships that characterise the inner workings of this design, an Entity- Relationship Model for the Airbnb database was developed to accommodate these requirements.
Entity-Relationship attributes such as hosts, guests and administrators were created with their suitable information such as name, email, phone number, address as well as personalised features such as wish lists for guests, saved payment methods are also included to maintain personalisation of the users. These attributes and entities were created with PostgresSQL(pgAdmin - 9.1) and datatypes mentioned in the Entity-Relationship Model are used accordingly.
Additionally, to enhance the browsing and booking reservations experience for the hosts and guests, a help-centre entity is also included.
Lastly, triple relationships such as hosts -> listing -> guests, were considered and utilised to ensure scalability. To document the variety of datatypes used and to maintain functionality for the attributes, data dictionary was prepared. These strategies and methods were selected to ensure that, the database will satisfy functional specifications and be sufficiently adaptable for future advancements.

5: References:

1. Airbnb | Holiday rentals, cabins, beach houses & more. (2024). Retrieved from Airbnb website: https://www.airbnb.co.in
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3. pgAdmin 4 — pgAdmin 4 7.0 documentation. (n.d.). Retrieved from www.pgadmin.org website: https://www.pgadmin.org/docs/pgadmin4/latest/index.html